22BCON1068

Assignment - 01

Life Skills - II (Aptitude)

DATE: 26/12/23
PAGE NO.: 1

Q1. What is the greatest five digit number that is completely divisible by 8, 15, 16, 21, and 5?

(i) 95760 (ii) 98320 (iii) 99120 (iv) 92680

Sol. LCM:

$$\begin{array}{r|l} 3 & 8, 15, 16, 21 \\ \hline 8 & 8, 5, 16, 7 \\ \hline & 1, 5, 2, 7 \end{array}$$

$3 \times 8 \times 5 \times 2 \times 7 = 1680$

Now, 95760 and 99120 both are divisible by 1680.

but greatest number = 99120

— (iii) Ans

Q2. The LCM of three different no. is 120. Which of the following can't be their HCF.

(a) 8 (b) 12 (c) 24 (d) 35

Sol. Here LCM = 120

So,

$$\begin{array}{r|l} 2 & 120 \\ \hline 2 & 60 \\ \hline 2 & 30 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$120 = 2 \times 2 \times 2 \times 3 \times 5$

$120 = 4(2 \times 3 \times 5)$

So, 4 is the common factor of three no. it mean HCF of all three no. will be multiple of 4.

Now, 35 is not multiple of 4.

So, Ans = 35

— (d)

Q3. The LCM of two no. is 4 times their HCF. The sum of LCM and HCF is 125. If one of the no. is 100. then the other no. is

(a) 5 (b) 25 (c) 100 (d) 125

Dec 26, 2023, 23:41

Dec 26, 2023, 23:38

22BCON1068

DATE: / /
PAGE NO.: 2

Sol. Let HCF = a

LCM = 4 × a = 4a

Now. Sum of LCM and HCF = 125

4a + a = 125

5a = 125

a = 25

So. HCF = 25

and LCM = 4a = 100

We know, HCF × LCM = Product of two number

Let second no. = x

HCF × LCM = 100 × x

25 × 100 = 100 × x

x = 25

So 2nd number = 25

— (b)

Q4 The product of two no. is 4107. if the HCF of the no. is 37. then a greater no. is

(a) 185 (b) 111 (c) 107 (d) 101

Sol HCF of two number = 37

Let. LCM = x

HCF × LCM = Product of two no.

$37 \times x = 4107$

$x = \frac{4107}{37}$

$x = 111$

— (b) Ans

Q5. Find the least no. which when divided seperately by 15, 20, 36 & 48 leaves 3 as reminder in each case.

(a) 183 (b) 243 (c) 483 (d) 723

Dec 26, 2023, 23:38

22BCON1068

DATE: / /

PAGE NO.: 3

Sol^n. LCM

3	15, 20, 36, 48
4	5, 20, 12, 16
5	5, ~~5~~, 3, 4
	1, 1, 3, 4

LCM = $3 \times 4 \times 5 \times 3 \times 4 = 720$

Reminder = 3

so

Least no. = $720 + 3 = 723$ Ans

— (d)